

APHTHOUS ULCERATION

(APTHAE, CANKER SORE,

RECURRENT APHTHOUS

ULCERATION)

Aphthous ulceration is an acute, recurrent, and painful condition characterized by oral ulcers that occur on nonkeratinized mucosa. It is the most common cause of oral ulcerations. Up to 30% of the population report experiencing recurrent aphthous ulcers, with most cases developing by age 30. One-third of children under 18 years also report having oral ulcers.

Aphthous ulcers typically present with a gray or yellow pseudomembrane surrounded by an erythematous margin. There are three recognized clinical types, as outlined in **Table 74-4**:

TABLE 74-4 Clinical Classification of Oral Aphthous Ulcers

Type	Clinical Appearance	Prevalence	Clinical Course
Minor	One or more < 5-mm shallow ulcers	90%–95%	7–10 days; resolve without scarring
Major	One or more 1- to 3-cm deep ulcers	5%–10%	~6 weeks; may scar
Herpetiform	1–100, 1- to 3-mm shallow ulcers	1%–5%	< 7 days

Etiology and Pathogenesis

Potential triggers include: - Heredity - Food or medication allergies - Decreased mucosal barrier integrity - Hematologic or immunologic disorders - Emotional stress - Trauma

Pregnancy is often associated with clinical improvement, while smoking cessation and use of sodium lauryl sulfate (common in toothpastes) may exacerbate symptoms.

The pathogenesis involves a cell-mediated immune response, with histologic findings showing a mononuclear infiltrate of T lymphocytes, macrophages, and mast cells, along with increased tumor necrosis factor (TNF).

Systemic Associations

Aphthous ulcers can be an oral manifestation of systemic conditions such as: - Acute HIV infection - Behçet disease - Reactive arthritis syndrome

Classification: Simple vs. Complex Aphthosis

- **Simple aphthosis:** One to three ulcers recurring one to three times per year.
- **Complex aphthosis:** Continuous ulceration (new lesions form before old ones heal), often associated with systemic disease or genital ulcers.

Evaluation and Diagnosis

A diagnostic workup may be indicated to rule out underlying systemic disease. A biopsy is recommended if: - The lesion fails to heal within 3–4 weeks - The clinical history is concerning

Histopathology typically shows nonspecific ulcerated mucosa with a chronic mixed inflammatory infiltrate.

Management

Topical therapies focus on supportive care and avoiding exacerbating factors.

Systemic treatments may include: - Prednisone - Colchicine - Thalidomide

Behçet Disease

Behçet disease (see Chap. 167) is a rare multisystem disorder most prevalent in Japan, Turkey, and the Middle East. It is characterized by: - Recurrent oral

ulcers (nearly universal) - Genital ulcers - Iridocyclitis - Variable involvement of synovitis, meningoencephalitis, and cutaneous pustular vasculitis

Diagnosis requires oral aphthae plus at least two of the other findings. Oral lesions resemble aphthous ulcers, heal within weeks, and recur at least three times per year.